

GSOE9210 Engineering Decisions

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Updating belief

- 1 Decisions and information
 - Airline case study
 - Experiments, observables, and information
 - Updating beliefs about likelihoods
 - Value of information
 - Updating reliability likelihood
- 2 Sensitivity analysis

Updating belief

1 Decisions and information

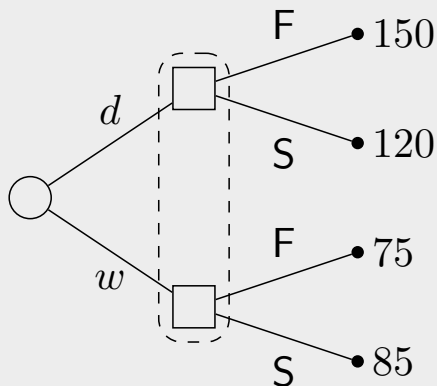
- Airline case study
- Experiments, observables, and information
- Updating beliefs about likelihoods
- Value of information
 - Updating reliability likelihood

2 Sensitivity analysis

Representing information

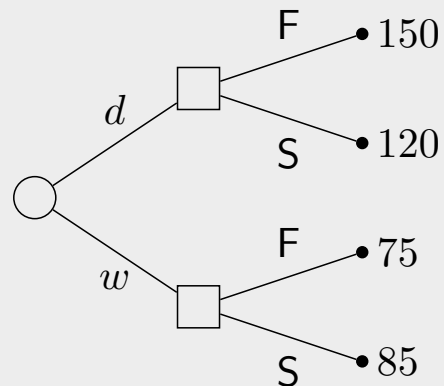
Consider the fund-raiser example.

- Decision before weather known:



- Decision nodes part of the same *information set*
- Possible choices: F, S only

- Decision after weather known:



- Decision nodes distinguishable
- Possible choices: d/F , d/S , w/F , w/S

Case study: capital purchase



Example (To purchase or not)

You're the chief engineer of a small commercial airline which, due to increased demand, is considering adding to its fleet by buying (B) a used plane. Another company is offering to sell one of its planes for \$400,000. Used planes range in reliability, which is hard to determine without a detailed inspection.

Question: should you purchase?

Problem modelling

- Problem 1: how to measure reliability? Operating hours
- Simplification 1: classify planes as either: very reliable (vR) ($> 90\%$), moderately reliable (mR), or unreliable (uR) ($< 50\%$)

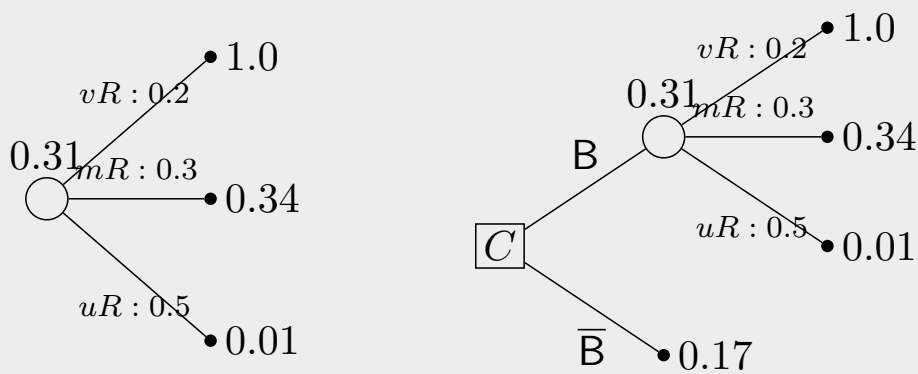
Beliefs about reliability:

	Reliability		
	vR	mR	uR
Probability	0.2	0.3	0.5
Utility	1.0	0.34	0.01

- Simplification 2: current operating profit \$170K (don't buy); very reliable plane increases profit to \$1M (best outcome); an unreliable one makes no profit (worst)
- Simplification 3: utility proportional to \$ ($u \propto v$)

Decision C (buy or not)

Buy:



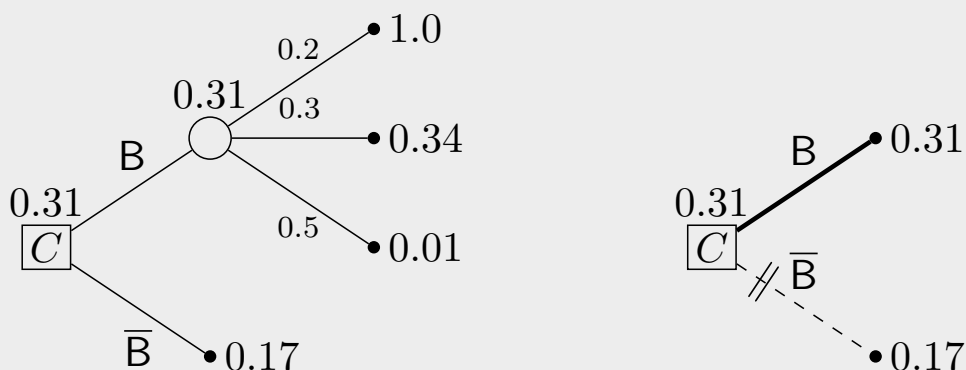
$$U(B) = 0.2(1.0) + 0.3(0.34) + 0.5(0.01) = 0.31$$

$$U(\bar{B}) = 0.17$$

	0.2	0.3	0.5	
	vR	mR	uR	U
B	1.00	0.34	0.01	0.31
$\bar{B}$	0.17	0.17	0.17	0.17

Decision C

- Evaluate decision points/nodes by maximal utility of alternatives (*i.e.*, actions/strategies)
- The value of node C is 0.31, because $0.31 > 0.17$; *i.e.*, $0.31 = \max\{0.17, 0.31\}$



Get more information?

Example (Additional information)

You have the option to consult an aeronautical engineering firm to conduct an assessment of the plane for \$10K. The assessment will include a *recommendation* as to whether to purchase: favourable (f) or unfavourable (u)

- Firm's assessment reliable?
- Records: 90% of very reliable planes receive favourable assessment; *i.e.*, $P(f|vR) = 0.9$

Probability of:	... conditional on:		
	vR	mR	uR
f	0.9	0.6	0.1
u	0.1	0.4	0.9

Observables and states

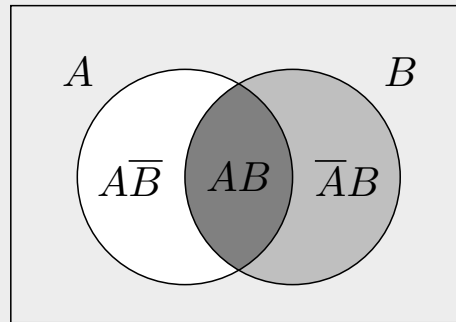
- Outcome depends on states: vR , mR , and uR
- Decision can only be based on *observables*: f and u
- Distinction between:
 - states**: determine outcomes but aren't usually directly observable
 - observations**: can be observed but aren't always reliable indicators of state

Definition (Experiments)

Experiments are actions which allow observations (*i.e.*, determine observables) about the state.

- Question: How do observables depend on states?
How do states depend on observables?

Conditional probability



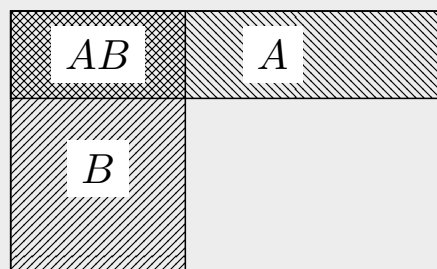
Definition

The *conditional probability* of event A conditional on B (provided B is possible; i.e., $P(B) \neq 0$), written $P(A|B)$, is defined by:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

In the diagram above, $P(A|B)$ represents the ratio of (the area of) the region AB (the dark region) to that of the whole of B .

Conditional independence



Definition

Event A is (*conditionally*) *independent* of event B if:

$$P(A|B) = P(A).$$

Event A is (*conditionally*) *dependent* on B if A is not (conditionally) independent of B .

For example, if B is a random sample of a population.

Bayes's rule

- Rearranging the definition of conditional probability:

$$P(A \cap B) = P(A|B)P(B)$$

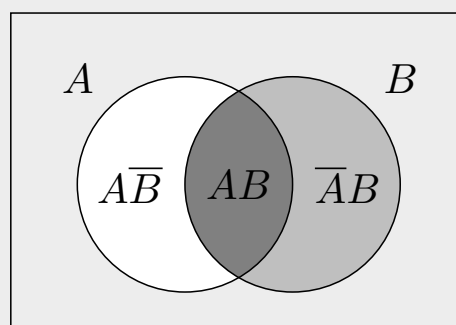
- By symmetry $P(A \cap B) = P(B \cap A)$;
therefore: $P(A|B)P(B) = P(B|A)P(A)$.
Rearranging gives:

Theorem (Bayes's Theorem I)

If A and B are any two events (such that $P(A) \neq 0$), then:

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

Bayes's Venn diagram



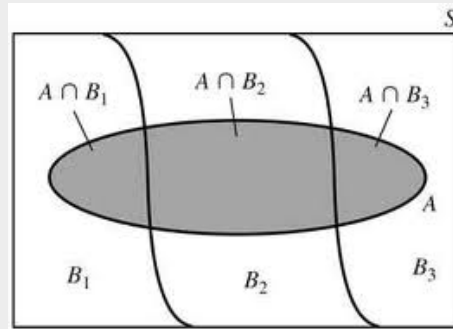
But $A = AB \cup A\bar{B}$. So we get the following:

Theorem (Bayes's Theorem I')

If A and B are any two events ($P(A) \neq 0$), then:

$$P(B|A) = \frac{P(AB)}{P(AB) + P(A\bar{B})}$$

Extending Bayes's rule



$$P(B_1|A) = \frac{P(AB_1)}{P(AB_1) + P(AB_2) + P(AB_3)}$$

$$P(B_2|A) = \frac{P(AB_2)}{P(AB_1) + P(AB_2) + P(AB_3)}$$

$$P(B_3|A) = \frac{P(AB_3)}{P(AB_1) + P(AB_2) + P(AB_3)}$$

Bayes's rule generalised

Events $B_1, \dots, B_n$ are *universally exhaustive* (of Ω) if $\bigcup_{i=1}^n B_i = \Omega$.

Theorem (Bayes's Theorem II)

If events $B_1, \dots, B_k, \dots, B_n$ are mutually exclusive and universally exhaustive, and A is a possible event ($P(A) \neq 0$), then:

$$P(B_k|A) = \frac{P(AB_k)}{\sum_{i=1}^n P(AB_i)}$$

Theorem (Bayes's Theorem II')

If $B_1, \dots, B_k, \dots, B_n$ are mutually exclusive and universally exhaustive events and A is a possible event ($P(A) \neq 0$) then:

$$P(B_k|A) = \frac{P(A|B_k)P(B_k)}{\sum_{i=1}^n P(A|B_i)P(B_i)}$$

Example: Bayes's rule

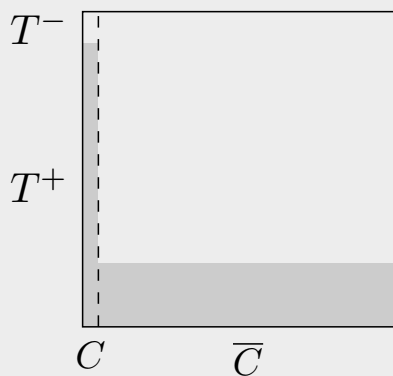
Example (Medical diagnostics)

In a given population of people, one in every thousand have hypo-cytocitic cancer. A certain pathology test is used to detect the disease. The test is 'good' but not perfect; it returns a positive result in 98% of persons with the disease, and registers a *false positive* (i.e., gives a positive result for a person free of the disease) 5% of the time.

Exercise

A random person comes in to get tested and the test returns a positive result. What is the probability that the person has cancer?

Example: solution



Given information:

$$\begin{aligned} P(C) &= \frac{1}{1000} & P(\bar{C}) &= \frac{999}{1000} \\ P(T^+|C) &= \frac{98}{100} & P(\bar{T}^+|C) &= \frac{2}{100} \\ P(T^+|\bar{C}) &= \frac{5}{100} & P(\bar{T}^+|\bar{C}) &= \frac{95}{100} \end{aligned}$$

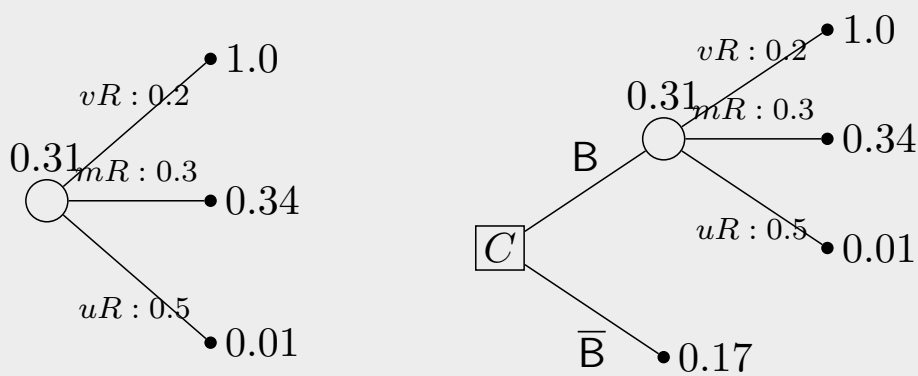
What is $P(C|T^+)$?

$$\begin{aligned} P(C|T^+) &= \frac{P(CT^+)}{P(T^+)} = \frac{P(CT^+)}{P(CT^+ \cup \bar{C}T^+)} = \frac{P(CT^+)}{P(CT^+) + P(\bar{C}T^+)} \\ &= \frac{P(T^+|C)P(C)}{P(T^+|C)P(C) + P(T^+|\bar{C})P(\bar{C})} \\ &= \frac{\frac{98}{100} \times \frac{1}{1000}}{\frac{98}{100} \times \frac{1}{1000} + \frac{5}{100} \times \frac{999}{1000}} = \frac{98}{98 + 5 \times 999} \approx \frac{100}{5000} = 0.02 \end{aligned}$$

Patient only has 2% chance of having cancer despite testing positive?!

Decision C (buy or not)

Buy:



$$U(B) = 0.2(1.0) + 0.3(0.34) + 0.5(0.01) = 0.31$$

$$U(\bar{B}) = 0.17$$

	0.2	0.3	0.5	
	vR	mR	uR	U
B	1.00	0.34	0.01	0.31
$\bar{B}$	0.17	0.17	0.17	0.17

Post-report (posterior) probabilities

- If report favourable (f):

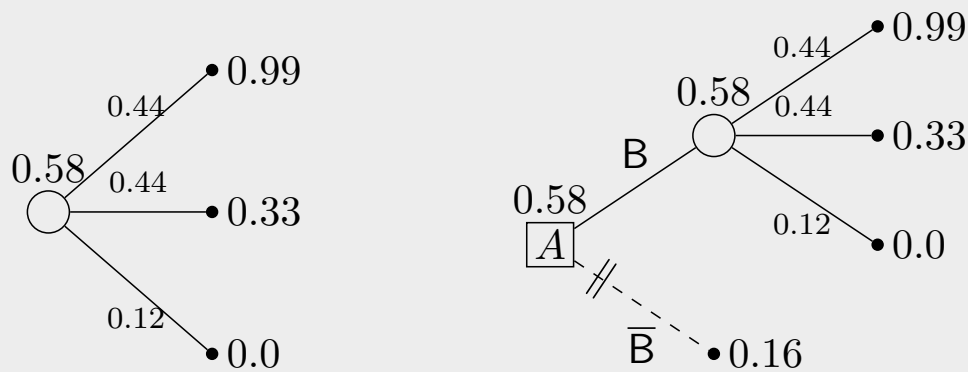
$$\begin{aligned}
 P(vR|f) &= \frac{P(f|vR)P(vR)}{P(f|vR)P(vR) + P(f|mR)P(mR) + P(f|uR)P(uR)} \\
 &= \frac{0.9(0.2)}{0.9(0.2) + 0.6(0.3) + 0.1(0.5)} \\
 &= \frac{0.18}{0.41} \approx 0.44
 \end{aligned}$$

Similarly: $P(mR|f) \approx 0.44$ and $P(uR|f) \approx 0.12$

- If report unfavourable (u):

$$\begin{aligned}
 P(vR|u) &= \frac{0.02}{0.59} \approx 0.04 \\
 P(mR|u) &\approx 0.20 \\
 P(uR|u) &\approx 0.76
 \end{aligned}$$

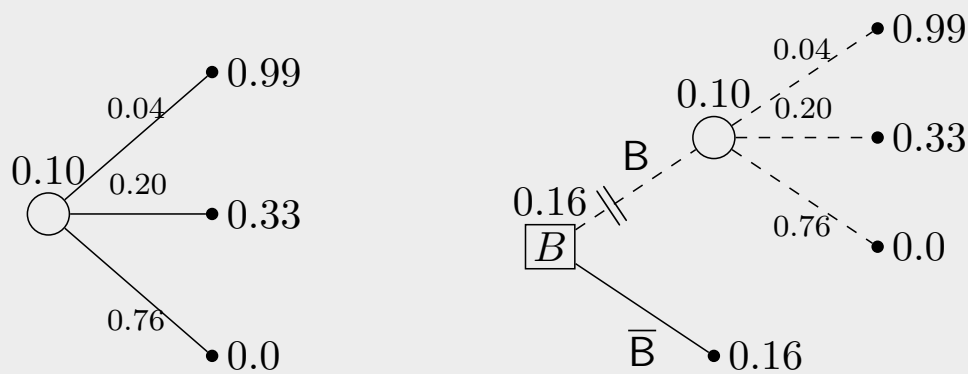
Decision A (report favourable)



- The revised expected utility of buying the plane is $U(B) = 0.44(0.99) + 0.44(0.33) + 0.12(0.0) = 0.58$
- The utility of not buying it is $U(\bar{B}) = 0.16$.

	0.44	0.44	0.12	
	vR	mR	uR	U
B	0.99	0.33	0.0	0.58
$\bar{B}$	0.16	0.16	0.16	0.16

Decision B (report unfavourable)



- The revised expected utility of buying the plane is $U(B) = 0.04(0.99) + 0.20(0.33) + 0.76(0.0) = 0.10$
- The utility of not buying it is $U(\bar{B}) = 0.16$.

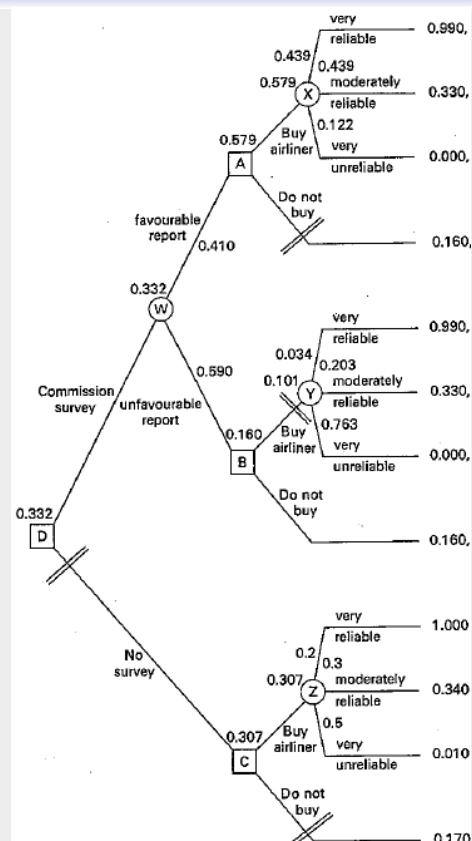
	0.04	0.20	0.76	
	vR	mR	uR	U
B	0.99	0.33	0.0	0.10
$\bar{B}$	0.16	0.16	0.16	0.16

Utility adjustments

- Problem: cost associated with report?
Question: How does report's cost (\$10K) affect utility?
- Observation: report cost small relative to other monetary quantities: potential profit \$1M; *i.e.*, $\$10K \ll \$1M$
- Simplification 4: model effect by constant shift; *i.e.*, for report costing \$ x ($x \ll 1M$), change of utility is $\Delta u = \frac{x}{1M}$;
- That is, \$10K is approximately worth 0.01 *utils*

Combined decision

- Combine all three possible cases into one big decision problem
- Introduce new decision: commission survey/report, and no survey/report
- Introduce new observable: report outcome (f or u)
- If consultant good, report likely to be good predictor of (*i.e.*, correlated to) aircraft reliability
- Consultant's increased predictive accuracy is *valuable* in making decision



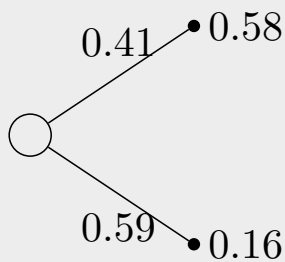
Combined decision

- From the denominators in the earlier calculations:

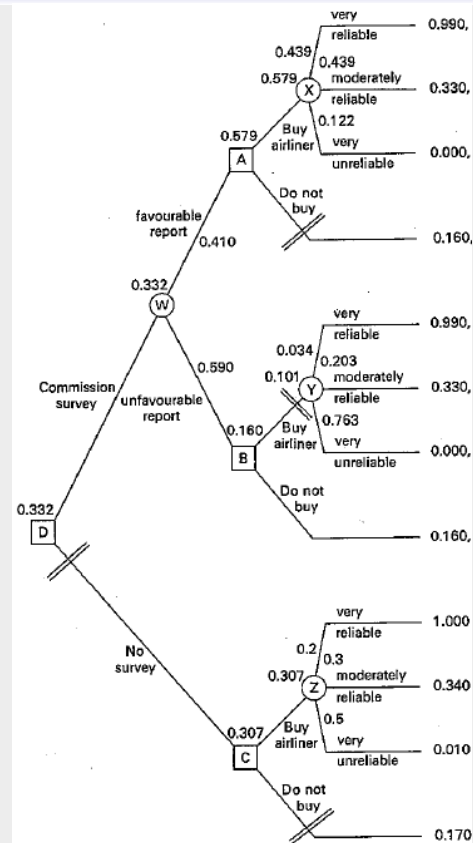
$$P(f) = 0.41$$

$$P(u) = 0.59$$

- Therefore, if report commissioned:



- Utility of report: 0.33



Decision table

	f, vR	f, mR	f, uR	u, vR	u, mR	u, uR	U
A_1	1.0	0.34	0.01	1.0	0.34	0.01	0.31
A_2	0.17	0.17	0.17	0.17	0.17	0.17	0.17
A_3	0.99	0.33	0	0.99	0.33	0	
$\vdots$	$\vdots$	$\vdots$		$\dots$			
A_6							

where

A_1 no survey; buy plane

A_2 no survey; don't buy plane

A_3 commission survey; buy plane

A_4 commission survey; don't buy

A_5 commission survey; if favourable, buy plane; else don't buy

A_6 commission survey; if favourable, don't buy plane; else buy

Value of information

- Optimal policy if report commissioned:

Policy for R: report commissioned

If report favourable, buy plane, if not don't buy.

- Value of policy is $U(R) = 0.33$, inclusive of the 0.01 fee
- Optimal policy if report not commissioned:

Policy for $\bar{R}$: report not commissioned

Buy the plane.

- $U(\bar{R}) = 0.31$
- How much is report worth?
- $U(R) = 0.34 - u_r \geq 0.31 = U(\bar{R})$; i.e., should commission report for fee up to $u_r = 0.03$; i.e., for any fee up to \$30K

Summary

- Explored decision problems in greater depth:
 - actions that affect epistemic state (value of information-gathering actions)
 - dealing with uncertainty in preferences (sensitivity analysis)
- Updating beliefs (epistemic state) via Bayes's theorem
- Value of information: cost of gathering more information versus increase in expected utility due to new information

Updating belief

1 Decisions and information

- Airline case study
- Experiments, observables, and information
- Updating beliefs about likelihoods
- Value of information
 - Updating reliability likelihood

2 Sensitivity analysis

Production and demand

Example (Production)

Alice is the CTO at a company and Bob is the CFO. They're considering two possible production processes for a product. Process A is expected to net \$40K if demand increases, \$30K if demand remains stable, and \$20K if demand falls. Process B requires a greater initial capital expenditure; it will only net \$10K if demand drops, and \$40K otherwise.

Future estimates of demand are: 20% of an increase, 30% chance of staying level, and 50% of a decrease.

Which process should Alice implement?

Example

The decision table is:

	$\frac{5}{10}$ ↓	$\frac{3}{10}$ —	$\frac{2}{10}$ ↑	$V_{\$}$
A	\$20	\$30	\$40	\$27
B	\$10	\$40	\$40	\$25

$$V_{\$}(A) = \frac{5}{10}(20) + \frac{3}{10}(30) + \frac{2}{10}(40)$$

$$= 10 + 9 + 8 = \$27$$

$$V_{\$}(B) = \frac{5}{10}(10) + \frac{3}{10}(40) + \frac{2}{10}(40)$$

$$= 5 + 12 + 8 = \$25$$

Alternative A has greater expected monetary value

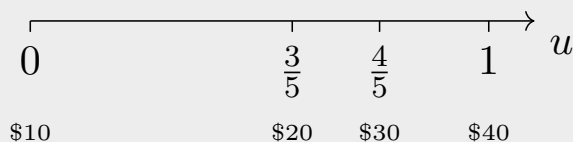
Example

Alice consults Bob who advises her that, under its current financial position, the company's preferences are:

$$\$20 \sim \left[\frac{3}{5} : \$40 \mid \frac{2}{5} : \$10 \right]$$

$$\$30 \sim \left[\frac{4}{5} : \$40 \mid \frac{1}{5} : \$10 \right]$$

The company's utility for money is:



The utility table:

	$\frac{5}{10}$ ↓	$\frac{3}{10}$ —	$\frac{2}{10}$ ↑	U
A	$\frac{3}{5}$	$\frac{4}{5}$	1	$\frac{74}{100}$
B	0	1	1	$\frac{50}{100}$

$$U(A) = \frac{5}{10}\left(\frac{3}{5}\right) + \frac{3}{10}\left(\frac{4}{5}\right) + \frac{2}{10}(1)$$

$$= \frac{1}{50}(15 + 12 + 10) = \frac{74}{100}$$

$$U(B) = \frac{5}{10}(0) + \frac{3}{10}(1) + \frac{2}{10}(1)$$

$$= \frac{1}{50}(0 + 15 + 10) = \frac{50}{100}$$

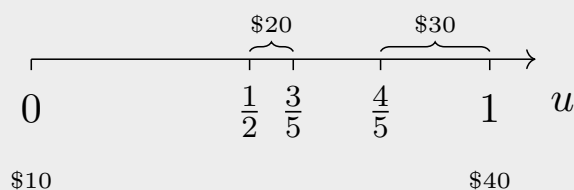
Therefore, A will also have greater utility

Sensitivity analysis

Suppose Bob cannot give precise assessments on values of \$20 and \$30, only bounds:

$$\begin{aligned} \left[\frac{3}{5}\$40\right] &\succ \$20 \succ \left[\frac{1}{2}\$40\right] \\ \$40 &\succ \$30 \succ \left[\frac{4}{5}\$40\right] \end{aligned}$$

The utility for money is:



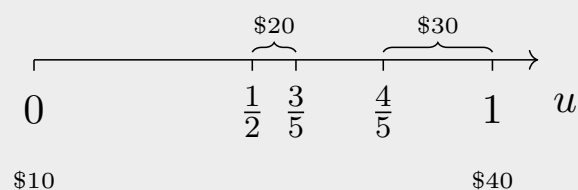
Lower bound for A:

	$\frac{5}{10}$ ↓	$\frac{3}{10}$ —	$\frac{2}{10}$ ↑	U
A	$\frac{1}{2}$	$\frac{4}{5}$	1	$\frac{69}{100}$
B	0	1	1	$\frac{50}{100}$

Upper bound for A:

	$\frac{5}{10}$ ↓	$\frac{3}{10}$ —	$\frac{2}{10}$ ↑	U
A	$\frac{3}{5}$	1	1	$\frac{80}{100}$
B	0	1	1	$\frac{50}{100}$

Sensitivity analysis



Bounds on A:

$$\begin{aligned} U(A) &> \frac{5}{10}\left(\frac{1}{2}\right) + \frac{3}{10}\left(\frac{4}{5}\right) + \frac{2}{10}(1) \\ &= \frac{1}{100}(25 + 24 + 20) \\ &= \frac{69}{100} \\ U(A) &< \frac{5}{10}\left(\frac{3}{5}\right) + \frac{3}{10}(1) + \frac{2}{10}(1) \\ &= \frac{1}{100}(30 + 30 + 20) \\ &= \frac{80}{100} \end{aligned}$$

That is:

$$\frac{69}{100} < U(A) < \frac{80}{100}$$

Conclusion:

A is guaranteed to be preferred to B ($U(B) = \frac{50}{100}$) regardless of the uncertainty over the precise preference for \$20 and \$30.

Summary

Sensitivity analysis:

- decisions under imprecise preferences
- how might preference uncertainty affect a decision?
- can be applied to anything which may be imprecise: payoffs, probabilities, *etc.*